Synopsis

Project Title: PulseTech – Pediatric Pharmacovigilance Dashboard using Heterogeneous Graph Attention Networks (HANConv) for Proactive Adverse Drug Reaction Detection

Problem Statement (In Brief): Design and develop an automated pediatric pharmacovigilance dashboard that combines retrospective Disproportionality Analysis (PRR, ROR, IC, EBGM) with a proactive Heterogeneous Graph Attention Network (HANConv) to detect known and predict unreported Adverse Drug Reactions (ADRs) in children aged 0–17, achieving at least 85% predictive accuracy (ROC-AUC) and reducing manual pharmacovigilance report preparation time from weeks to under 60 seconds per report.

Problem Description / Problem Analysis

Background & Context:

• Adverse Drug Reactions (ADRs) are the 4th leading cause of death in the United States, and children face disproportionately higher risk because their metabolic profiles differ drastically from adults. Over 50% of pediatric prescriptions are administered 'off-label' – meaning the drug was never formally tested or approved for use in children. The U.S. FDA's Adverse Event Reporting System (FAERS) collects millions of spontaneous safety reports, but the publicly available tools only provide simple, retrospective case counts without any statistical significance testing or predictive capability.

Current Situation:

• **What is the problem?** Existing pharmacovigilance systems (including the FDA Public Dashboard) are purely reactive – they only identify drug safety signals after a statistically large number of children have already been harmed. They provide raw event counts but do not compute whether those counts represent a mathematically significant anomaly (disproportionality). Furthermore, they have zero capability to predict ADRs for newly introduced drugs that lack historical reports.

• **Who is affected?** Pediatric patients (Neonates 0–27 days, Infants 28 days–2 years, Children 2–11 years, Adolescents 12–17 years), clinical pharmacists responsible for drug safety monitoring, pharmaceutical companies conducting post-market surveillance, and regulatory bodies (FDA, EMA, WHO-UMC).

• **Scale of problem:** The FDA FAERS database contains over 25 million reports (2021–2025). Approximately 8–12% of these involve pediatric patients. Manual analysis of even a single drug class takes a pharmacovigilance team 2–4 weeks using Excel/SAS, costing organizations ,000–,000 per report cycle. Meanwhile, children continue to be exposed to undetected off-label risks.

Quantifiable Impact:

• Time wasted: 2–4 weeks per drug class report (manual CSV download, data cleaning in Excel/SAS, formula computation, and Word/PDF formatting).

• Errors/failures: Raw FAERS data contains >30% misspelled drug names and inconsistent brand/generic mappings, leading to fragmented and missed signals.

• Cost implications: Industry-standard pharmacovigilance report generation costs ,000–,000 per cycle per drug class.

• Safety concerns: Delayed detection of pediatric ADRs has historically led to drug withdrawals (e.g., Cisapride, Terfenadine) only after significant pediatric morbidity and mortality.

• Statistical evidence: Studies show that traditional case-counting misses ~35% of true signals that are detectable only through disproportionality analysis (Evans et al., 2001).

Existing Solutions and Limitations:

• FDA FAERS Public Dashboard: Provides raw case counts only. No statistical signal detection (PRR/ROR/IC/EBGM), no predictive modeling, no automated reporting. Users must manually download CSVs.

• OpenVigil / VigiBase (WHO): Academic/regulatory tools that compute some DPA metrics but lack graph-based predictive intelligence and are not pediatric-specific.

• Commercial Tools (Argus Safety, ArisGlobal): Extremely expensive (–/year licenses), require dedicated teams, and still lack GNN-based prediction capabilities.

• Why can't we use them? None offer the dual-engine approach (retrospective DPA + proactive GNN prediction) specifically tailored for pediatric age bands, and none provide one-click automated PDF report generation.

Scope of Proposed Solution:

• Technology/Approach: Python (FastAPI backend), PyTorch Geometric (HANConv GNN), PostgreSQL database, React.js frontend dashboard. Data standardized via RxNorm and MedDRA APIs.

• What the system includes: (a) Automated FAERS data ingestion pipeline, (b) Disproportionality Analysis engine (PRR, ROR, IC, EBGM), (c) HANConv GNN for predictive link detection, (d) Interactive dashboard with filtering by age group and drug, (e) One-click regulatory-grade PDF report generation.

• How it differs: It is the only system that combines gold-standard retrospective statistical validation with proactive graph-neural-network predictions – using a concept called 'Guilt by Association' to predict ADRs for drugs with zero or few historical reports, all within a pediatric-specific context.

Objectives

[Each measurable and achievable in 4 months]

1. To design and implement an automated data ingestion pipeline that parses raw FDA FAERS XML/CSV files and standardizes drug names via RxNorm API and adverse event terms via MedDRA, so that data quality issues are eliminated, achieving >95% mapping accuracy.

2. To develop a Disproportionality Analysis (DPA) engine that computes four gold-standard signal detection algorithms (PRR, ROR, IC, EBGM) with their confidence intervals, so that retrospective drug-ADR signals are confirmed with mathematical rigor.

3. To construct a heterogeneous graph from FAERS data and train a HANConv model using PyTorch Geometric, so that unreported Drug-ADR relationships can be predicted via link prediction with an ROC-AUC ≥ 0.85.

4. To build a full-stack web dashboard (React.js + FastAPI) with interactive visualizations, age-group filtering, and network graph exploration, so that pharmacovigilance professionals can explore safety signals in real-time.

5. To implement an automated PDF Report Builder that generates regulatory-grade safety reports in under 60 seconds per drug class, so that the manual 2–4 week report preparation cycle is eliminated.

Evaluation Criteria for Problem Statement

Use this checklist: [Filled by Team]

Criterion	Questions	Acceptable Level	Our Answer
Relevance	Is it connected to your field? Is it current?	Must align with your branch/courses	MCA – Data Science & Big Data Analytics. Uses Machine Learning (GNN), Web Development (React/FastAPI), and Database Management (PostgreSQL). Directly

			aligns with DS, DBMS, and ML coursework.
Feasibility	Can you complete it in 3–4 months? Do you have resources?	60–70% technical confidence	Duration: 4 months. Tech: Python, PyTorch Geometric, React.js, PostgreSQL. Infrastructure: Personal laptops with GPU support. Confidence: 80% – all libraries are mature and well-documented.
Novelty	Is it original or a meaningful improvement?	Doesn't need to be brand new, but add value	Meaningful innovation: First system to combine DPA statistical validation + HANConv GNN prediction in a pediatric-specific pharmacovigilance dashboard with automated PDF report generation.
Impact	Does it solve a real problem? Who benefits?	Clear end-user or stakeholder	End-users: Clinical pharmacists, pharmaceutical safety teams, regulatory bodies (FDA/EMA). Impact: Reduces report generation from weeks to seconds; predicts ADRs before children are harmed.
Complexity	Is it appropriately challenging?	Intermediate (not trivial, not research-level)	Challenge: Heterogeneous graph construction from messy FDA data, training HANConv with meta-path attention, integrating statistical engine + ML pipeline + full-stack dashboard into one cohesive system.
Scope	Not too large, not too small?	Can demonstrate 2–3 key features well	Key features: (1) DPA Signal Detection Engine, (2) HANConv GNN Predictor, (3) Interactive Dashboard with PDF Report Builder.
Resources	Tools, software, hardware accessible?	Use free/low-cost alternatives if needed	All free/open-source: Python, FastAPI, PyTorch Geometric, React.js, PostgreSQL, RxNorm API, MedDRA.

Team Details

Team No:

Name	SRN	Roll No	Email ID	Contact No
Atharv Telkar	SCFP125043	43	athervtelkar08@gmail.com	8788762307
Amey Telkar	SCFP125044	44	ameytelkar08@gmail.com	9834242475

Guide Name & Signature: _____